

A Calibrated PCA–Machine Learning Pipeline for Biosignature Candidate Triage in Exoplanet Transmission Spectra

Owen Yang^a

^a Henry M. Gunn High School, 780 Arastradero Rd, Palo Alto, CA 94306, USA.

owenhyang@gmail.com

Abstract

Characterizing exoplanetary atmospheres efficiently is essential for future survey missions such as Ariel, where only a subset of planets can receive the most detailed follow-up observations. Existing classification models have demonstrated great promise in performing this task; however, they have predominantly been trained on existing low-resolution spectra. With the Ariel mission anticipating high-dimensional spectra for classification, there is an urgent demand for models which can classify synthetic transmission spectra as biosignature or non-biosignature cases. We simulated synthetic spectra at a resolving power of $R = 200$ (550 spectral bins), matching the highest-precision benchmark mode (Tier 3) planned for the Ariel mission. We used MultiREx / TauREx 3, cleaned invalid spectra, and trained four supervised classifiers—XGBoost, Random Forest, Multi-Layer Perceptron, and 1D Convolutional Neural Network—on PCA-transformed spectral features. PCA was used to reduce dimensionality while a secondary standardization step was applied to equalize the variance between the dominant physical systematics of the first two components and the subtle, chemically relevant absorption features contained in higher-order components.

The gradient-boosted decision tree ensemble (XGBoost) achieved the strongest overall performance, with a top mean accuracy of 88.17% ($\pm 1.99\%$) and an F1-score of 88.24% ($\pm 2.22\%$). XGBoost decisively outperformed the Multi-Layer Perceptron (79.34% $\pm 1.47\%$) and the 1D-CNN (75.53% $\pm 1.05\%$), and held a narrower, paired-test edge over the Random Forest (86.51% $\pm 1.96\%$) that did not reach significance on an aggregate bootstrap comparison. Whereas the gradient-based neural networks required explicit whitening of the PCA feature space to remain competitive—and still trailed the ensembles—the scale-invariant, node-splitting architecture of XGBoost exploited the chemically informative low-variance components natively, isolating absorption features from the dominant physical baselines. Crucially, XGBoost produced the best-calibrated probability estimates among the models tested (Brier score: 0.0861, with a near-diagonal reliability curve), making its confidence estimates well suited to candidate-prioritization decisions, pending validation on more realistic atmospheric data. These results suggest that XGBoost can serve as an exceptionally robust, calibrated triage tool for prioritizing scarce telescope resources.

1. Introduction

With more than 6,000 confirmed exoplanets discovered to date and major NASA and ESA research programs now explicitly focused on planetary habitability and the search for life, the characterization of exoplanetary atmospheres has become a central problem in modern observational astrobiology (1, 2). Identifying exoplanets capable of sustaining life requires the robust detection of atmospheric biosignatures—gases whose presence strongly implies a biological origin, but only when interpreted within sufficient geological, atmospheric, and stellar context, since many candidate biosignature gases can be produced or mimicked abiotically (3, 4). In alignment with the primary science goals of the upcoming Ariel (5, 6) mission, the ultimate objective of this endeavor is not merely to identify individual planetary candidates, but to conduct a comprehensive chemical census of exoplanetary atmospheres.

The Ariel mission, launching in 2029, will survey approximately 1,000 transiting exoplanets using visible-to-infrared photometry and medium- and low-resolution infrared spectroscopy spanning 0.5–7.8 μm during a nominal four-year science mission (5). To optimize valuable telescope time, the mission employs a multi-tiered observation strategy: Tier 1 acts as a broad reconnaissance survey, Tier 2 conducts a deeper baseline chemical survey across hundreds of atmospheres, and Tier 3 is strictly reserved for the most detailed, high-resolution observations of exceptional targets (5, 7). Because the most detailed follow-up is highly resource-intensive, a rigorous triage process is required to rank the Tier 3 benchmark planets and determine which warrant dedicated Tier 4 phase-curve observations or follow-up with ground-based Extremely Large Telescopes (ELTs). By simulating data at $R=200$, this study develops a machine learning framework capable of serving as this critical intra-Tier 3 prioritization filter. However, without existing empirical high-resolution spectra of this caliber, machine learning models capable of serving as this intra-Tier-3 filter remain under active development, necessitating the use of synthetic data to build and evaluate these predictive algorithms in advance of the mission.

Existing exoplanet atmosphere studies have used transmission spectroscopy and high-resolution Doppler spectroscopy to detect molecular species such as H_2O , CO , and CH_4 , with many of the strongest detections coming from large, close-in giant planets whose extended atmospheres produce stronger observational signals (8, 9). However, extending these techniques to smaller, rocky, potentially habitable planets remains difficult because their atmospheric signals are weaker and more easily affected by noise, clouds, stellar contamination, and observational time limits (3, 10). High-resolution spectroscopy offers a promising pathway because it can resolve molecular absorption into many individual spectral lines, producing molecule-specific patterns that can be isolated through Doppler shifts and cross-correlation techniques (9, 11). This makes HRS useful for identifying trace atmospheric species and helping reduce degeneracies when combined with lower-resolution spectra, although it does not by itself confirm biological activity (3, 11). We emphasize that the present work operates in a different regime: the $R = 200$ spectra studied here represent the high-resolution end of Ariel's low-to-medium-resolution survey, not the $R \sim 10^4\text{--}10^5$ Doppler-resolved spectroscopy described above. Throughout, "high-resolution" is used relative to the $R \approx 10\text{--}100$ data used in prior biosignature-classification studies.

Although real exoplanet atmosphere spectra are publicly available, existing observations do not yet provide the kind of large, uniform, biosignature-labeled dataset needed for supervised biosignature candidate screening. The NASA Exoplanet Archive’s Atmospheric Spectroscopy Table compiles downloadable transmission, eclipse/emission, and direct-imaging spectra from peer-reviewed studies, but these observations vary across target type, facility, instrument, wavelength coverage, spectral resolution, and spectrum type (12). Much of the comparative broadband transmission-spectroscopy literature has focused on hot Jupiters and other large, highly irradiated planets because their stronger atmospheric signals make them easier to characterize; for example, Sing et al. (13) analyzed ten hot Jupiters across 0.3–5 μm , while the atmospheric properties of smaller rocky exoplanets remain largely unconstrained (14). While ground-based high-resolution spectroscopy has successfully detected molecules such as carbon monoxide in hot-Jupiter atmospheres, these observations are typically limited to favorable targets and do not yet provide broad coverage across planetary types or atmospheric compositions (9). Ariel will address this problem at the population level by obtaining a large, homogeneous survey of exoplanet atmospheres across a broad visible-to-infrared wavelength range, enabling comparative studies of atmospheric composition, clouds, and thermal structure (5, 7). Because Ariel-like surveys and synthetic spectral grids can be large and computationally demanding to analyze, they motivate the development of efficient machine-learning methods that can assist atmospheric retrieval and classification (15, 16).

In this work, we aim to design a machine learning framework capable of distinguishing between biosignature and non-biosignature environments using high-resolution (HR) transmission spectra from an Ariel-like dataset. To generate these data, we use the MultiREx library, an open-source Python framework that leverages the TauREx 3 radiative transfer code to rapidly simulate planetary transits and robustly model molecular absorption and scattering (17, 18). We process these high-dimensional spectra using a Principal Component Analysis (PCA) pipeline to isolate relevant chemical signals. In evaluating our framework, we emphasize balanced detection performance and probability calibration rather than any single metric, while noting that the operating threshold can later be tuned to prioritize recall—minimizing the risk of discarding potentially habitable worlds—at deployment (Section 4.5). This is supported by an extensive analysis of model error rates across diverse physical parameter spaces (e.g., planetary radius and stellar temperature).

The closest precedent for this work is Duque-Castaño et al. (17), who used MultiREx-generated synthetic spectra to classify low-SNR, low-resolution transmission spectra of Earth-like/TRAPPIST-1e-style planets as biosignature candidates, framing the task as triage for follow-up rather than full retrieval. We adopt their MultiREx-based generation, their CH_4/O_3 chemistry, and their candidate-screening framing, and extend that foundation along three axes that prior biosignature-classification work has largely left unexplored. First, we move from the low-to-moderate-resolution regime ($R \approx 10\text{--}100$) used in earlier studies (17, 19) to a simulated resolving power of $R = 200$ representative of Ariel’s Tier 3 benchmark, where each spectrum spans hundreds of correlated wavelength channels and dimensionality reduction becomes essential rather than optional. Second, we run a controlled, like-for-like comparison of four

architectures spanning two families—two tree-based ensembles (Random Forest and gradient-boosted XGBoost) and two neural networks (an MLP and a 1D-CNN)—on a single shared PCA-derived feature space, isolating the effect of model family from that of preprocessing. Third, we treat probability calibration as a first-class evaluation criterion alongside accuracy, since a triage tool's confidence estimates—not only its labels—drive resource-allocation decisions, yet calibration is rarely reported in this setting. We note that the relative robustness of tree ensembles over neural networks on PCA-projected tabular features is consistent with a well-established pattern in machine learning; our contribution is not that result in isolation but its combination with explicit calibration and a physics-based error analysis showing that failures concentrate in physically expected regimes (low scale height, near-threshold abundances). Together these establish a calibrated, interpretable benchmark for biosignature-pair screening at Ariel Tier-3 resolution, with validation on real, heterogeneous spectra as the clear next step.

In this paper, Section 2 outlines the state of existing machine learning research for exoplanet habitability and makes the case for high-resolution ensemble classification, Section 3 describes the synthetic dataset, the PCA-based feature extraction, the four model architectures, and the validation framework; Section 4 presents the results, including a rigorous statistical comparison using McNemar's tests and a physics-based error analysis. Finally, Section 5 concludes the paper by summarizing key findings and outlining avenues for future research.

2. Background

Previous work on exoplanet atmospheric analysis has often focused on atmospheric retrieval, where models infer physical and chemical properties such as molecular abundances, cloud opacity, and temperature from observed spectra. Traditional Bayesian retrieval methods are powerful because they produce posterior probability distributions rather than simple labels, but they can also be computationally expensive, especially when the parameter space is large or highly degenerate. Márquez-Neila et al. (19) showed that supervised machine learning could reduce this computational burden by training a Random Forest model on a precomputed grid of atmospheric models. Their approach retrieved atmospheric parameters from exoplanet spectra more quickly than standard retrieval methods while preserving some interpretability through feature importance. However, this type of retrieval-focused method is designed to estimate atmospheric parameters, not necessarily to serve as a rapid binary screening tool for prioritizing biosignature follow-up observations.

More recent studies have explored deep learning and other machine learning approaches for exoplanet spectra, especially in the context of Ariel-like datasets. Yip et al. (20) investigated how deep learning models can be interpreted when applied to exoplanet atmospheric retrievals, highlighting both their predictive power and the concern that neural networks often function as “black boxes.” Their work is relevant to this study because it shows that model evaluation should go beyond accuracy alone: researchers must also understand when models fail, which spectral regions influence predictions, and whether predictions are physically meaningful. Similarly, the

Ariel Data Challenge and related studies have emphasized that machine learning can help analyze large synthetic spectral datasets, but many of these approaches focus on retrieval or posterior estimation rather than the narrower task of identifying spectra that deserve follow-up observation. This interpretability concern is directly relevant to the present work and is discussed further in Section 4.3, where we examine how model errors map onto physical parameter space as a means of partially validating the physical plausibility of model predictions.

The most directly related work is Duque-Castaño et al. (17), who developed a machine-learning framework for classifying low-SNR synthetic transmission spectra according to whether they contained potential biosignatures or bioindicators such as methane, ozone, and water. Their study used MultiREx to generate large numbers of synthetic spectra and framed machine learning as a way to identify “interesting” planets for follow-up, rather than as a replacement for detailed atmospheric retrieval. This is especially important because biosignature searches are limited by telescope time: a model that can quickly flag promising targets could make future observing campaigns more efficient. However, their study was limited to a specific Earth-like/TRAPPIST-1e-style setup using low-resolution spectra, and did not directly compare the same set of models across the broader parameter space or the higher-resolution regime anticipated for next-generation surveys such as Ariel. Building on these approaches, this project evaluates tree-based ensembles, neural networks, and PCA-based feature extraction as a supervised classification pipeline for synthetic transmission spectra, with particular attention to recall and error patterns across physical parameter space.

3. Methods

This study aims to design a machine learning model capable of distinguishing between biosignature and non-biosignature exoplanets using high-resolution transmission spectra from an Ariel-like dataset. We assess a suite of machine learning models on mock data simulated via the MultiREx library, using the following methodologies.

3.1 Dataset and simulation parameters

The synthetic dataset of atmospheric transmission spectra was generated using the MultiREx (v0.3.1) Python library, which leverages the TauREx 3 radiative transfer code (18). TauREx 3 is a modular framework designed to simulate the transit of starlight through a planetary atmosphere by integrating the Beer-Lambert law across discrete atmospheric layers. It incorporates a wide range of physical effects, including pressure-broadened molecular absorption, Rayleigh scattering, and collision-induced absorption (CIA), to produce physically consistent transmission spectra. While TauREx 3 is highly robust, its accuracy is fundamentally limited by the underlying molecular cross-section data and the assumption of a 1D, spherically symmetric atmosphere (18, 21). Furthermore, certain edge cases in the parameter space—such as atmospheres with extremely high base pressures or those with no significant absorption features—can occasionally result in numerical instability or physically implausible transit depths

(> 1.0). To maintain the integrity of the dataset, these invalid spectra were identified and removed during the data-cleaning phase described below.

The parameters for the dataset used in this study was drawn from the Ariel Big Challenge (Tier 2) introduced by Changeat and Yip (15), and the data was generated using the TauREx 3 radiative transfer code (18). We generated a training set of 2,696 simulated planetary spectra (reduced from 3,000 after removing spectra with physically impossible transit depths > 1.0 or NaN values), as well as five independent testing sets, each containing 540 clean spectra. Each sample contained a one-dimensional transmission spectrum (transit depth as a function of wavelength) with 550 spectral bins spanning wavelengths from 0.5 to 7.8 micrometers at a spectral resolution of $R = 200$, with a constant signal-to-noise ratio (SNR) of 15 applied to all observations, following the low-SNR transmission-spectrum framework explored by Duque-Castaño et al. (17). By adopting a uniform $R = 200$ grid, our synthetic dataset approximates the upper end of Ariel's Tier 3 benchmark resolution ($R \approx 200$ in AIRS-CH0, with lower native resolution in the other channels), allowing us to test prioritization strategies on data approaching the highest quality the mission will produce. This is an idealization: the real mission delivers heterogeneous resolution across its instruments, which we do not model here.

For supervised classification, binary labels were assigned based on the presence of a specific biosignature pair: methane and ozone, following the binary classification framework described by Duque-Castaño et al. (17). A spectrum was labeled as a “Biosignature” case only if both gases were present above predefined abundance thresholds:

Methane: $\log \text{mixing ratio} > -6.0$

Ozone: $\log \text{mixing ratio} > -7.0$

All planets that did not meet both conditions simultaneously were labeled as “Non-Biosignature.” In addition to the spectra, several physical parameters—including planetary radius, stellar temperature, and the abundances of other trace gases (e.g., H_2O , CO_2 , CO)—were simultaneously varied along the pre-defined grid outlined in Table 1.

To guarantee a balanced dataset (50% positive, 50% negative) and prevent majority-class bias, we enforced a stratified generation plan consisting of four disjoint composition profiles:

1. Biosignature Profile (50.0% of dataset): CH_4 is drawn log-uniformly from $[-6.0, -3.0]$, and O_3 is drawn log-uniformly from $[-7.0, -1.0]$ (both gases strictly above thresholds).
2. Non-Bio Methane Profile (16.66%): CH_4 is drawn from $[-6.0, -3.0]$, and O_3 is drawn from $[-10.0, -7.0]$ (methane only).
3. Non-Bio Ozone Profile (16.66%): CH_4 is drawn from $[-9.0, -6.0]$, and O_3 is drawn from $[-7.0, -1.0]$ (ozone only).
4. Non-Bio None Profile (16.68%): CH_4 is drawn from $[-9.0, -6.0]$, and O_3 is drawn from $[-10.0, -7.0]$ (both below thresholds).

By simulating these specific, disjoint profiles using the `explore_multiverse` function in MultiREx, we ensure that the overall abundance boundaries across all planets span the full [-9.0, -3.0] and [-10.0, -1.0] grids while maintaining a highly balanced, robust training split. All other physical parameters (planet radius, mass, stellar temperatures, and background gas abundances) were simultaneously drawn independently and uniformly from the absolute bounds outlined in Table 1 across all four profiles.

Table 1. Parameter ranges used to generate the synthetic transmission spectra for the training and test sets. The 'Minimum' and 'Maximum' columns give the sampling bounds for each parameter, and 'Grid Spacing' indicates whether values were drawn from a uniform or log-uniform distribution across that range. These bounds were chosen to span a broad slice of the Ariel target population—considerably wider than the TRAPPIST-1e-like terrestrial regime of Duque-Castaño et al. (17)—and are informed by the Ariel target samples of Edwards & Tinetti (7), Changeat et al. (22), and Changeat & Yip (15). The broad uniform-sampling strategy follows the grid-based approach used to build synthetic training sets for machine-learning retrieval (19), while the CH₄/O₃ biosignature chemistry and detection thresholds follow Duque-Castaño et al. (17). As described in Section 3.1, CH₄ and O₃ abundances were drawn using a stratified generation plan that enforces a balanced 50/50 class distribution, rather than independent uniform sampling across the full grid (which would otherwise yield ~33.3% positive cases).

Parameter	Minimum	Maximum	Grid Spacing
Planet Radius (Earth Radii)	1	26	uniform
Planet Masses (Earth masses)	1	300	uniform
Atmospheric Temperature (Kelvin)	500	2500	uniform
Star Temperature (Kelvin)	2500	7500	uniform
Star Radius (Solar Radii)	0.1	1.7	uniform
Star Mass (Solar Masses)	0.1	1.7	uniform
H ₂ O	-10.0	-1.0	log-uniform
CO	-9.0	-3.0	log-uniform

CO ₂	-9.0	-3.0	log-uniform
NH ₃	-9.0	-3.0	log-uniform
CH ₄	-9.0	-3.0	log-uniform
O ₃	-10.0	-1.0	log-uniform

3.2 Preprocessing and Feature Engineering

To account for differences in scale across planets and host stars, Z-score standardization was applied to the raw spectral flux values. Principal Component Analysis (PCA) was then used to reduce the dimensionality of the dataset. PCA is an orthogonal linear transformation that projects data into a new coordinate system via the eigendecomposition of the data's covariance matrix, allowing us to isolate distinct physical and chemical signals into uncorrelated components.

Following the PCA transformation, a secondary standardization step was applied to the retained components using a StandardScaler fit on the training set and applied to all test sets, rescaling the 102 components to zero mean and unit variance. This whitening step prevented the neural network architectures from being dominated by the high-variance physical baselines (PC0 and PC1) at the expense of the lower-variance but chemically critical higher-order components. Because tree-based models are invariant to monotonic per-feature scaling, this step left XGBoost and Random Forest unaffected while equalizing the feature space across all four architectures. An ablation confirms the importance of this step: removing it lowers neural-network accuracy substantially (MLP 79.34% $\rightarrow$ 74.81%; 1D-CNN 75.53% $\rightarrow$ 64.51%), while the scale-invariant tree models are unaffected (Section 4.1).

The final feature set retained the first 102 principal components. These components preserve the dominant variation in the spectra while reducing computational cost and limiting overfitting. Because transmission spectra contain both broad physical structure and localized molecular absorption features, retaining the leading principal components allows the models to use the full spectral context available in the simulated observations.

To validate that these 102 components are sufficient to capture the chemical signal, the figure illustrates a PCA reconstruction and residual error across wavelengths for a representative test spectrum. The top panel demonstrates that a spectrum reconstructed using these principal components captures all major molecular absorption features. The bottom panel confirms this, showing a flat, low-magnitude fractional residual error centered at zero, indicating that the chosen number of components provides a near-lossless compression of the diagnostic chemical fingerprints.

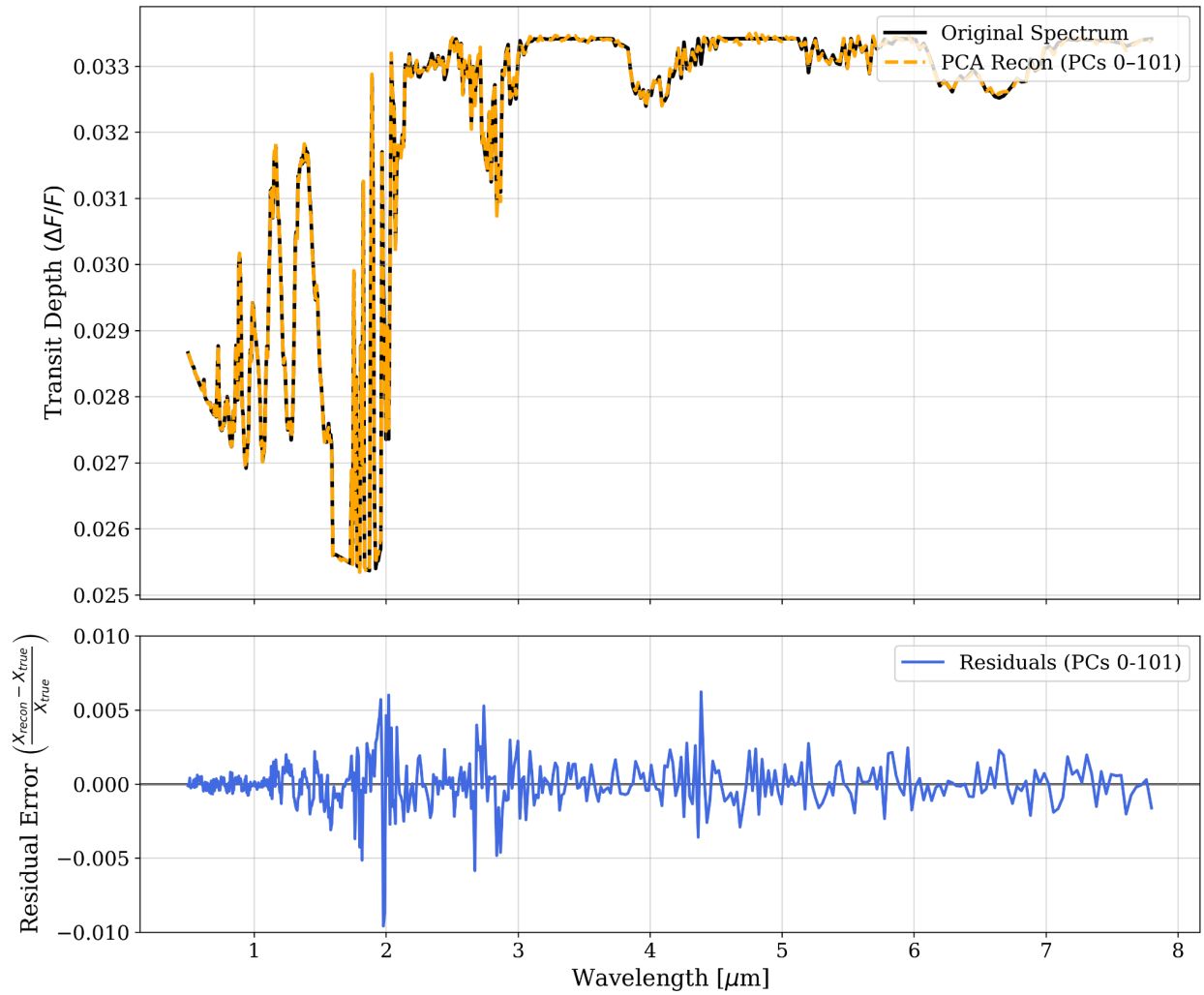


Figure 1: PCA Reconstruction Fidelity and Residual Analysis. The top panel displays the ground-truth synthetic spectrum (blue) for a representative H_2 -dominated exoplanet overlaid with the reconstruction utilizing 102 principal components (PCs 0–101, orange dashed). The bottom panel shows the fractional residual error, defined as $(X_{\text{recon}} - X_{\text{true}})/X_{\text{true}}$. The near-zero residuals and precise alignment of absorption peaks confirm that the selected components preserve all diagnostic chemical fingerprints while providing efficient dimensionality reduction.

3.3 Machine Learning Architectures

Two families of machine learning models—tree-based ensembles and neural networks—were evaluated, spanning four architectures to determine which approach was best suited for this spectral classification task. These models were selected to cover a spectrum of complexity, from robust ensemble methods capable of handling non-linear decision boundaries to deep neural networks that can automatically learn hierarchical feature representations.

Tree-based ensembles were included as established nonlinear supervised-learning baselines: Random Forests are robust ensembles of decision trees, while XGBoost is a scalable

gradient-boosted tree method widely used in machine-learning applications (23, 24). Because Random Forest methods have also been applied to exoplanet atmospheric spectra, they provide a relevant benchmark for this type of spectral analysis (19). Prior work on exoplanet transmission spectra has shown that spectral data are highly correlated and can be represented using lower-dimensional principal-component representations, supporting the use of PCA-transformed inputs (25). An MLP was included as a fully connected neural-network baseline for learning nonlinear relationships among the PCA components (26). Finally, the 1D-CNN was included because convolutional models have been applied to spectroscopic exoplanet data to detect molecular patterns in the spectral dimension and handle challenging noise conditions (27).

3.3.1 Gradient Boosted Decision Trees (XGBoost)

An ensemble of decision trees was trained using the XGBoost framework. This approach was selected for its ability to model complex, non-linear relationships and to perform well on difficult cases near the detection thresholds (24). The grid search evaluated learning rates of 0.01, 0.05, 0.1, and 0.2, maximum tree depths of 3, 5, 7, and 10, estimator counts of 100, 200, and 300, and subsample ratios of 0.8 and 1.0. The optimal configuration was determined to be 200 estimators with a maximum tree depth of 5, a subsample ratio of 0.8, and a learning rate of 0.2, utilizing a standard log-loss objective function.

3.3.2 Random Forest

A Random Forest classifier was evaluated as an additional tree-based ensemble approach. Implemented using scikit-learn (28), this model constructs a multitude of decision trees during training, each on a random subset of the data and features, and outputs the consensus prediction. This approach is highly robust against overfitting and effectively captures complex, non-linear relationships in high-dimensional spectral data. The hyperparameters for the Random Forest model were optimized using an exhaustive grid search with stratified 3-fold cross-validation, encompassing 108 distinct combinations across estimator counts of 100, 200, and 300, maximum depths of 10, 20, 30, or unbounded (None), minimum sample requirements of 2, 5, or 10 for node splitting, and 1, 2, or 4 samples per leaf. The model maximized performance using 300 decision trees with unbounded depth, requiring at least 2 samples per leaf and 2 samples per internal node split.

Table 2. Hyperparameter search ranges for the tree-based ensemble models. Bold values indicate the optimal configuration selected by grid search. Equivalent search ranges for the MLP and 1D-CNN architectures are described in Sections 3.3.3 and 3.3.4 respectively.

Model	Hyperparameter	Range
XGBoost	Learning Rate	{0.01, 0.05, 0.1, 0.2 }
	Estimators	{100, 200 , 300}
	Maximum Depth	{3, 5 , 7, 10}
	Subsample Ratio	{ 0.8 , 1.0}

Random Forest	Estimators Maximum Depth Minimum Samples Split Minimum Samples Leaf	{100,200, 300 } { None , 10, 20, 30} { 2 , 5, 10} {1, 2 , 4}

3.3.3 Deep Multi-Layer Perceptrons (MLP)

Training was performed for up to 100 epochs and controlled dynamically using early stopping with a patience of 10 epochs based on validation loss. The MLP grid evaluated hidden layer configurations of 256-128, 256-128-64, and 512-256-128-64 neurons, dropout rates within 0.2 and 0.4, batch sizes within 32, 64, and 128, and learning rates of 0.001 and 0.0005. The winning configuration utilized a three-layer funnel with 256, 128, and 64 neurons, respectively. Each dense layer was followed by Batch Normalization to stabilize training, a ReLU activation function, and a 0.3 dropout rate to mitigate overfitting (26). The network was optimized using the Adam algorithm with a learning rate of 0.001 and trained in batches of 128 (29).

3.3.4 Convolutional Neural Networks (1D-CNN)

To explore end-to-end learning directly from spectra, convolutional neural networks were initially trained on the raw, non-PCA spectral data. Both standard 1D CNNs and deeper 1D ResNet-style architectures with residual connections were tested. The standard 1D-CNN consisted of two convolutional layers with filters of 32 and 64 respectively. Each convolutional layer was followed by batch normalization, ReLU activation, 1D max pooling, and dropout ($p = 0.3$). To evaluate deeper feature extraction, a ResNet architecture was implemented employing residual blocks with skip connections (32) to mitigate the vanishing gradient problem, utilizing a Gaussian smoothing step (kernel size 3) prior to scaling to reduce high-frequency noise. Ultimately, models trained directly on the raw high-dimensional spectra failed to converge on generalizable features, yielding performance akin to random guessing (~50% accuracy). However, when the 1D-CNN was instead trained on the PCA-transformed feature space (specifically components 0-101), performance improved dramatically, validating the necessity of dimensionality reduction for datasets of this size.

3.4 Validation Framework

Model performance was evaluated using a hybrid validation approach dictated by the data requirements of the respective architectures. For the tree-based ensemble models (XGBoost and Random Forest), hyperparameters were systematically optimized using an exhaustive grid search with stratified 3-fold cross-validation. A 3-fold split was specifically chosen over higher fold counts (e.g., 10-fold CV) to balance robust internal validation while maintaining a sufficiently large training set per fold. However, initial cross-validation trials with the deep neural networks (MLP and 1D-CNN) revealed severe underfitting due to data starvation when fragmenting the relatively small dataset (2,696 samples). Maximizing the size of the training set was critical for stable deep learning performance. Therefore, the neural network architectures were optimized

and evaluated using a single-split validation approach, with 80% of the data reserved for training and 20% for validation and early stopping.

To assess robustness, five disjoint, independently generated test sets (~540 spectra each, with no overlap with the training set or with one another) were evaluated; the PCA basis and StandardScaler were fit on the training set only and applied unchanged to each. Each model was evaluated on all five sets, and we report the mean and standard deviation of four key metrics, Accuracy, Precision, Recall, and F1-Score in Section 4. Finally, a physics-based analysis was conducted by examining how classification errors varied with physical parameters such as stellar temperature and gas abundance, allowing the identification of regions in parameter space where the models were most likely to fail.

All data generation, modeling, and evaluations were performed under a unified software environment using Python v3.10.6. The synthetic spectra were generated using MultiREx v0.3.1 built on the TauREx v3.3.2 radiative transfer engine. The tree-based ensembles were developed using scikit-learn v1.7.2 (for Random Forest) and XGBoost v3.2.0. The deep learning architectures (MLP and 1D-CNN) were compiled and trained using TensorFlow v2.21.0 and scikeras v0.13.0 wrappers. To guarantee absolute algorithmic reproducibility across all partitions, initializations, and cross-validation folds, a global seed lock was enforced throughout (SEED = 42).

4. Results and Discussion

4.1 Overall Performance and Model Hierarchy

To ensure a fair comparison, all four architectures were trained and evaluated using the unified 102-component feature space (indices 0–101). Under these identical conditions, the gradient-boosted decision tree ensemble (XGBoost) demonstrated the strongest overall performance, achieving a top mean accuracy of 88.17% ($\pm 1.99\%$) and an F1-score of 88.24% ($\pm 2.22\%$). XGBoost decisively outperformed the Multi-Layer Perceptron (79.34% $\pm 1.47\%$) and the 1D-CNN (75.53% $\pm 1.05\%$), and held a narrower, paired-test edge over the Random Forest (86.51% $\pm 1.96\%$) that did not reach significance on an aggregate bootstrap comparison. We attribute this gap to the way each model family handles the multi-scale variance structure of the PCA representation. As Section 4.2 shows, the first two components carry 98.4% of the variance yet are essentially uncorrelated with the biosignature label, while the chemically informative signal resides in the low-variance tail. Tree-based models, whose node-splitting is invariant to per-feature scale, exploit those low-variance components natively, with no rescaling required. The neural networks cannot: without the secondary per-component standardization, they are dominated by the high-variance physical baselines and their accuracy falls to 74.81% (MLP) and 64.51% (1D-CNN). The whitening step rescues them by placing all components on a common scale—raising the MLP to 79.34% and the 1D-CNN to 75.53%—but does not close the gap to the trees. This residual deficit is consistent with the broader finding that tree ensembles outperform neural networks on structured, medium-sized tabular feature spaces (30, 31), and is likely compounded by the modest size of the training set (2,696 samples), which constrained the

neural architectures (Section 3.4). The trees' scale-invariance thus makes them robust to a variance structure that the neural networks can accommodate only with explicit whitening, and even then less effectively.

Model	Accuracy	Precision	Recall	F1-Score
XGBoost	88.17% ($\pm$ 1.99%)	88.02% ($\pm$ 1.30%)	88.50% ($\pm$ 3.45%)	88.24% ($\pm$ 2.22%)
Random Forest	86.51% ($\pm$ 1.96%)	87.17% ($\pm$ 2.28%)	85.85% ($\pm$ 2.74%)	86.48% ($\pm$ 2.06%)
MLP	79.34% ($\pm$ 1.47%)	72.30% ($\pm$ 1.52%)	95.57% ($\pm$ 1.46%)	82.32% ($\pm$ 1.29%)
1D-CNN	75.53% ($\pm$ 1.05%)	68.72% ($\pm$ 0.79%)	94.33% ($\pm$ 1.61%)	79.51% ($\pm$ 0.76%)
MLP (unwhitened)	74.81% ($\pm$ 2.61%)	83.83% ($\pm$ 3.11%)	61.80% ($\pm$ 3.42%)	71.14% ($\pm$ 3.35%)
1D-CNN (unwhitened)	64.51% ($\pm$ 0.79%)	58.97% ($\pm$ 0.66%)	96.91% ($\pm$ 0.84%)	73.32% ($\pm$ 0.48%)
1D-CNN (Raw)	50.56% ($\pm$ 1.58%)	50.42% ($\pm$ 1.01%)	85.01% ($\pm$ 2.00%)	63.29% ($\pm$ 1.25%)

Table 3. Mean $\pm$ standard deviation of performance metrics across the five independent test sets.

To rigorously evaluate the statistical significance of these results, we employed a dual-validation approach. First, pairwise McNemar's tests (33) were conducted on the models' predictions to evaluate paired classification agreement. These tests confirmed that the gradient-boosted decision trees (XGBoost) maintain a highly statistically significant predictive advantage over both the Random Forest model ($p = 0.0084$) and the MLP ($p = 0.0002$). While the Random Forest and MLP are statistically similar to each other ($p = 0.0818$), both architectures demonstrated a highly statistically significant advantage over the 1D-CNN ($p < 0.0001$), confirming the CNN's position at the bottom of the predictive hierarchy in this high-resolution feature space. However, to verify that these findings remain robust when evaluating the balance of precision and recall across the aggregate population, we performed a bootstrapping analysis (34) with 10,000 iterations to calculate the 95% confidence intervals for the F1-scores. This aggregate analysis revealed that the F1-score confidence intervals of the two top-performing ensembles (XGBoost and Random Forest) exhibit an overlap, with the F1-score difference being statistically non-significant at the population level ($p = 0.1924$). (“Website,” n.d.)

This divergence between McNemar's paired test (which showed highly significant case-by-case XGBoost superiority at $p = 0.0084$) and the bootstrap F1 test is a known statistical phenomenon. While the bootstrap test operates on aggregate, unpaired metrics and introduces sampling variance that dilutes minor differences, McNemar's test evaluates paired case-by-case disagreements. This indicates that, although the aggregate population F1-scores are statistically close, on a planet-by-planet basis XGBoost's predictive edge over Random Forest is systematic and statistically significant ($p < 0.01$); we therefore treat XGBoost as the most robust candidate for mission triage. (We note that these pairwise comparisons are not corrected for multiple testing, and that the XGBoost–Random Forest advantage—statistically significant on the paired McNemar test but not on the aggregate bootstrap F1 comparison—is therefore best read as suggestive rather than decisive.)

Across five independent test sets, XGBoost achieved a mean accuracy of 88.17% ($\pm 1.99\%$), while the MLP reached 79.34% ($\pm 1.47\%$). A representative confusion matrix for the XGBoost model is presented in Figure 2, illustrating strong discriminative performance on one independent test set. Across all five test sets, XGBoost achieved a mean recall of 88.50% ($\pm 3.45\%$) and an F1-score of 88.24% ($\pm 2.22\%$).

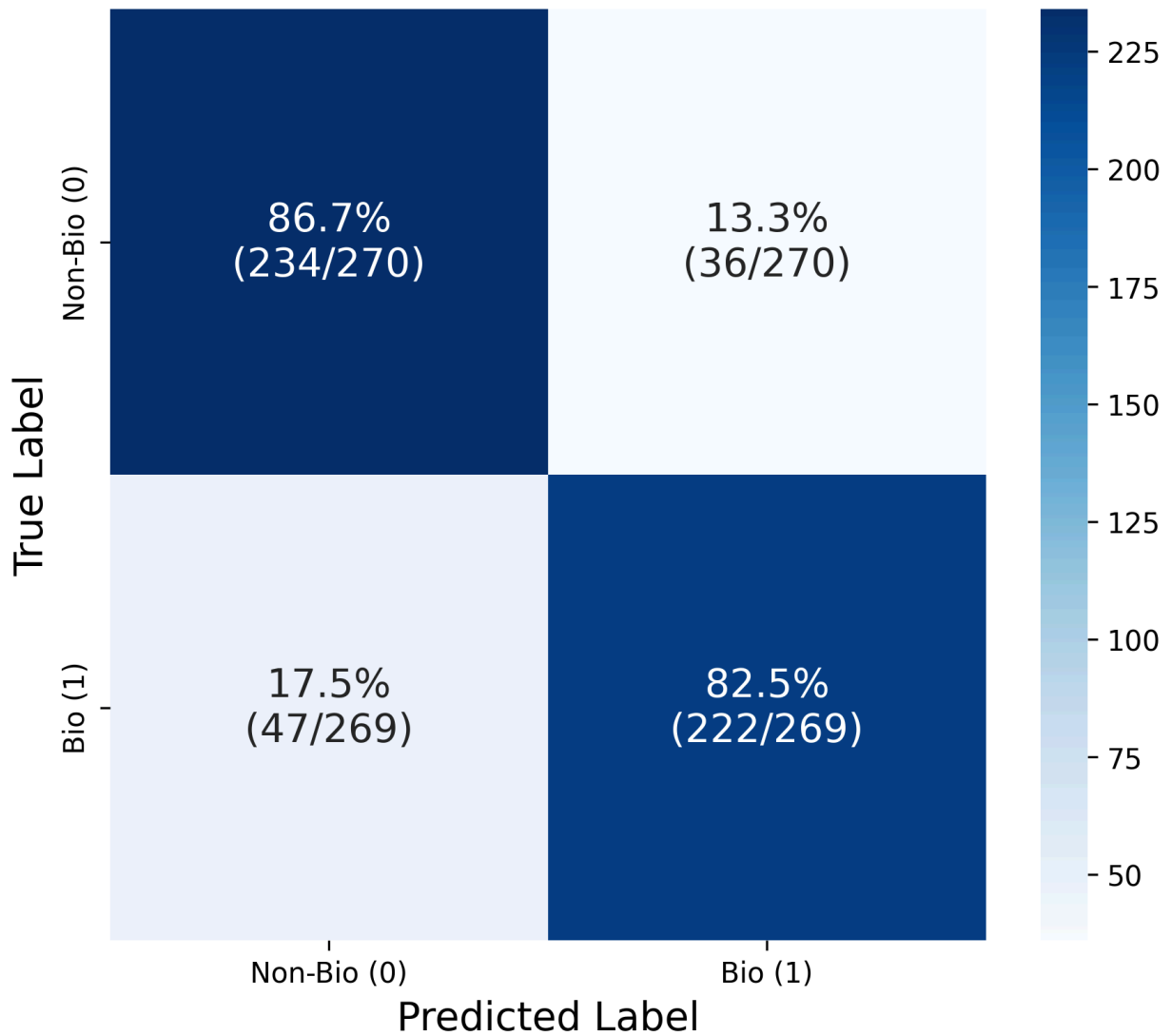


Figure 2: Confusion matrix for the best-performing XGBoost model evaluated on an independent test set. Accuracy values are displayed as percentages, with raw counts and relative fractions provided for each quadrant. The results demonstrate strong discriminative performance with controlled Type I (False Positive) and Type II (False Negative) errors, though the model shows a slight preference for high recall to minimize missed biosignature detections.

4.2 The Critical Role of PCA in Feature Extraction

A central finding of this research is the undeniable importance of Principal Component Analysis (PCA) as both a dimensionality reduction and denoising step. While the best-performing models utilized PCA-transformed data, the 1D-CNN trained on raw spectral data performed significantly worse, with the raw CNN peaking at only 50.56% ($\pm 1.58\%$) accuracy across the 5 independent test sets.

The contribution of PCA can be decomposed into two distinct steps. Trained on the raw 550-bin spectra, the 1D-CNN failed to learn generalizable features, reaching only 50.56% ($\pm 1.58\%$)

accuracy—barely above chance. Projecting onto the 102 principal components raised it to 64.51% ($\pm 0.79\%$), and the subsequent per-component whitening raised it further to 75.53% ($\pm 1.05\%$). The same whitening step lifted the MLP from 74.81% ($\pm 2.61\%$) to 79.34% ($\pm 1.47\%$). Dimensionality reduction and variance equalization thus contribute separately to neural-network performance, while the scale-invariant tree models—unaffected by either standardization—already operate near their ceiling on the PCA features.

Quantitative correlation analysis confirms that the first two principal components (PC0 and PC1) are physically tied to broad systematics rather than chemical features. Although these components explain approximately 98.41% of the total spectral variance, they showed practically zero quantitative correlation with the target biosignature labels (Pearson correlation coefficients of $r = 0.0075$ and $r = -0.0235$ respectively). By analyzing the component loadings and reconstructing the spectra, we determined that PC0 acts as a nearly perfect proxy for the mean transit depth ($r = 0.9998$) and PC1 captures the overall spectral slope. Thus, they encode broad physical properties like planetary radius and stellar continuum level rather than specific, high-frequency chemical absorption features. Meanwhile, Principal Components 2 through 101 successfully capture these vital high-frequency chemical absorption features.

The final feature set was constructed using 102 principal components (indices 0–101). The first 102 principal components cumulatively explain approximately 100% of the variance, with PC0 and PC1 accounting for 98.408% and PCs 2–101 accounting for the remaining $\sim 1.592\%$.

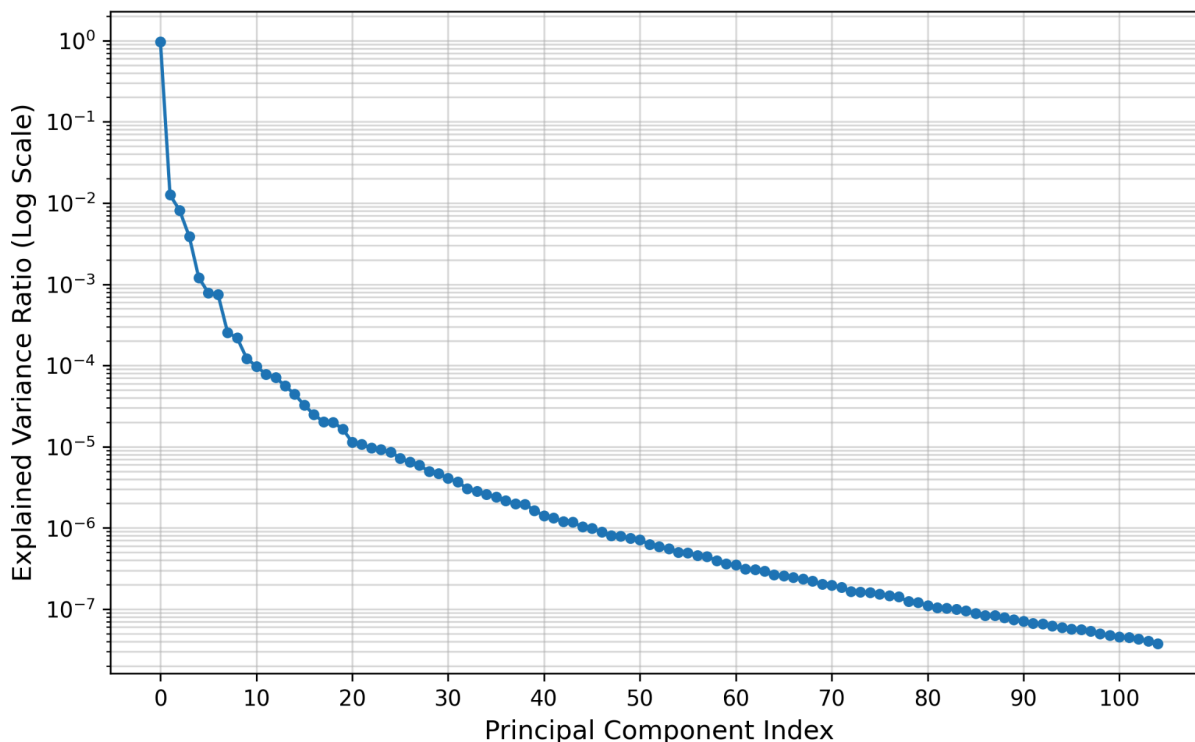


Figure 3: PCA Scree Plot illustrating the explained variance ratio for the first 105 principal components on a logarithmic scale. The first two components (PC0 and PC1) account for

98.408% of the total variance but capture broad physical systematics. The next 100 components (indices 2–101) capture approximately 1.592% of the variance, representing the high-frequency chemical absorption features required for biosignature classification.

4.3 Error Analysis and Physical Dependencies

To understand model failure modes, a physics-based analysis was conducted by examining how classification errors varied with physical parameters. A summary of the parameter ranges explored is provided in Table 1.

Figure 4 presents a triangle plot of planet and stellar parameters (Planet Radius, Planet Mass, Star Temp), with data points colored based on correct versus incorrect test set predictions.

Furthermore, Figure 5 provides a similar triangle plot focusing on atmospheric parameters (mixing ratios of CH₄ and O₃).

To understand where our models succeed and fail, a physics-based error analysis was conducted by mapping classification errors against physical planetary parameters (visualized in Figure 4 and Figure 5). The models demonstrated high accuracy for inflated, hot planets with prominent atmospheres. However, predictive performance degraded significantly in two specific physical regimes. First, the models struggled with planets possessing low atmospheric temperatures and small radii (and thus high surface gravities). This aligns perfectly with physical intuition: cooler temperatures and higher gravities compress the atmospheric scale height, which flattens the transmission spectrum and mutes the amplitude of diagnostic chemical absorption features beneath the noise floor. Second, as seen in Figure 5, misclassifications clustered heavily around the predefined biosignature boundaries ($\text{Log}(\text{CH}_4) \sim -6.0$ and $\text{Log}(\text{O}_3) \sim -7.0$). This highlights the inherent difficulty of classifying marginal, low-abundance detections in low-SNR (SNR = 15) data, where atmospheric signals are easily masked by simulated instrument noise.

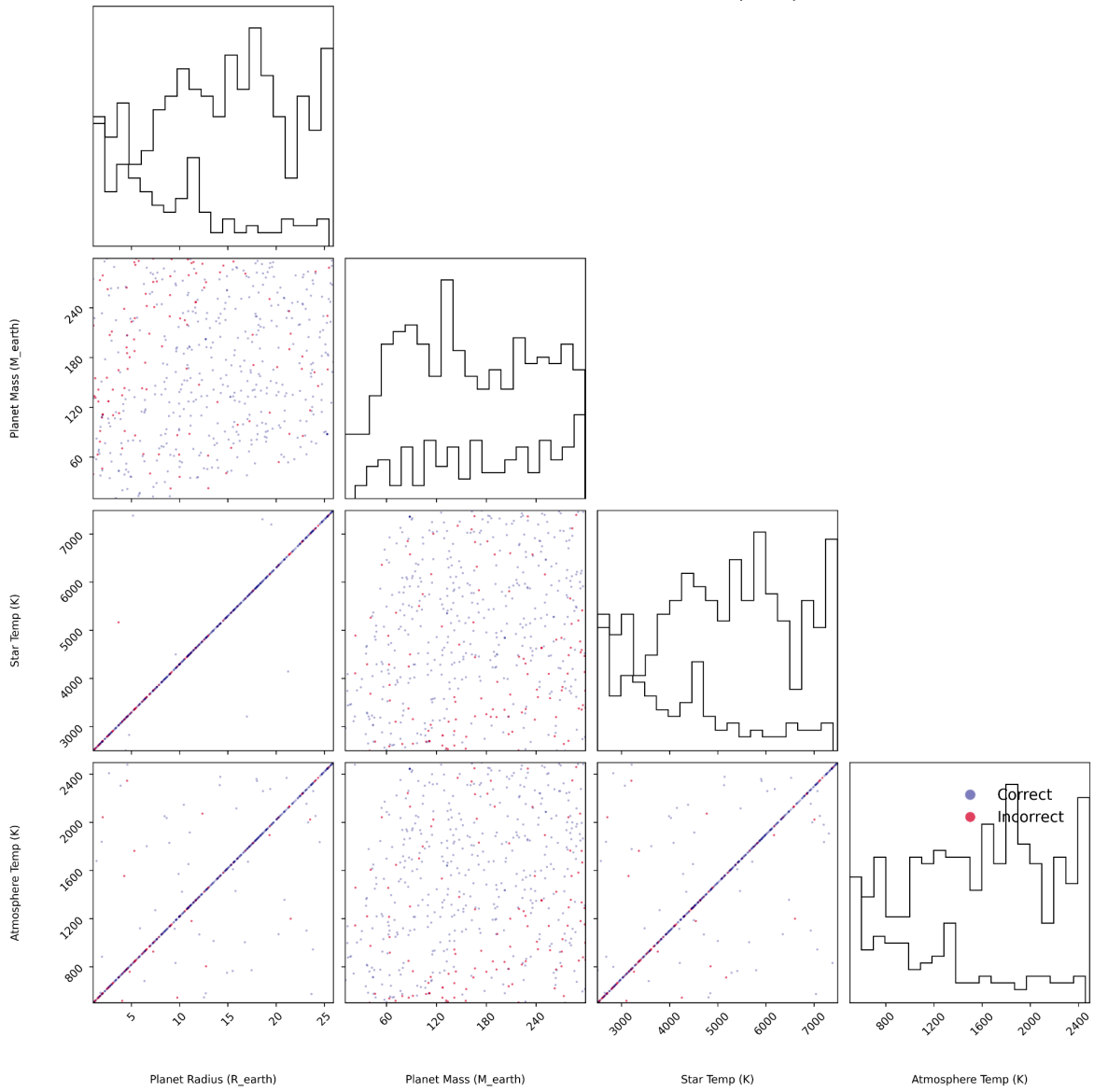


Figure 4: Scatter corner plot of classification errors across physical planetary and stellar parameters for the XGBoost model. Blue dots represent correct classifications, while red dots indicate errors. A distinct clustering of errors is visible in the regime of small planetary radii ($R_p < 8 R_{\oplus}$) and low atmospheric temperatures ($T < 1000$ K), where compressed atmospheric scale heights result in spectral signatures that fall below the simulated instrument noise floor ($\text{SNR} = 15$).

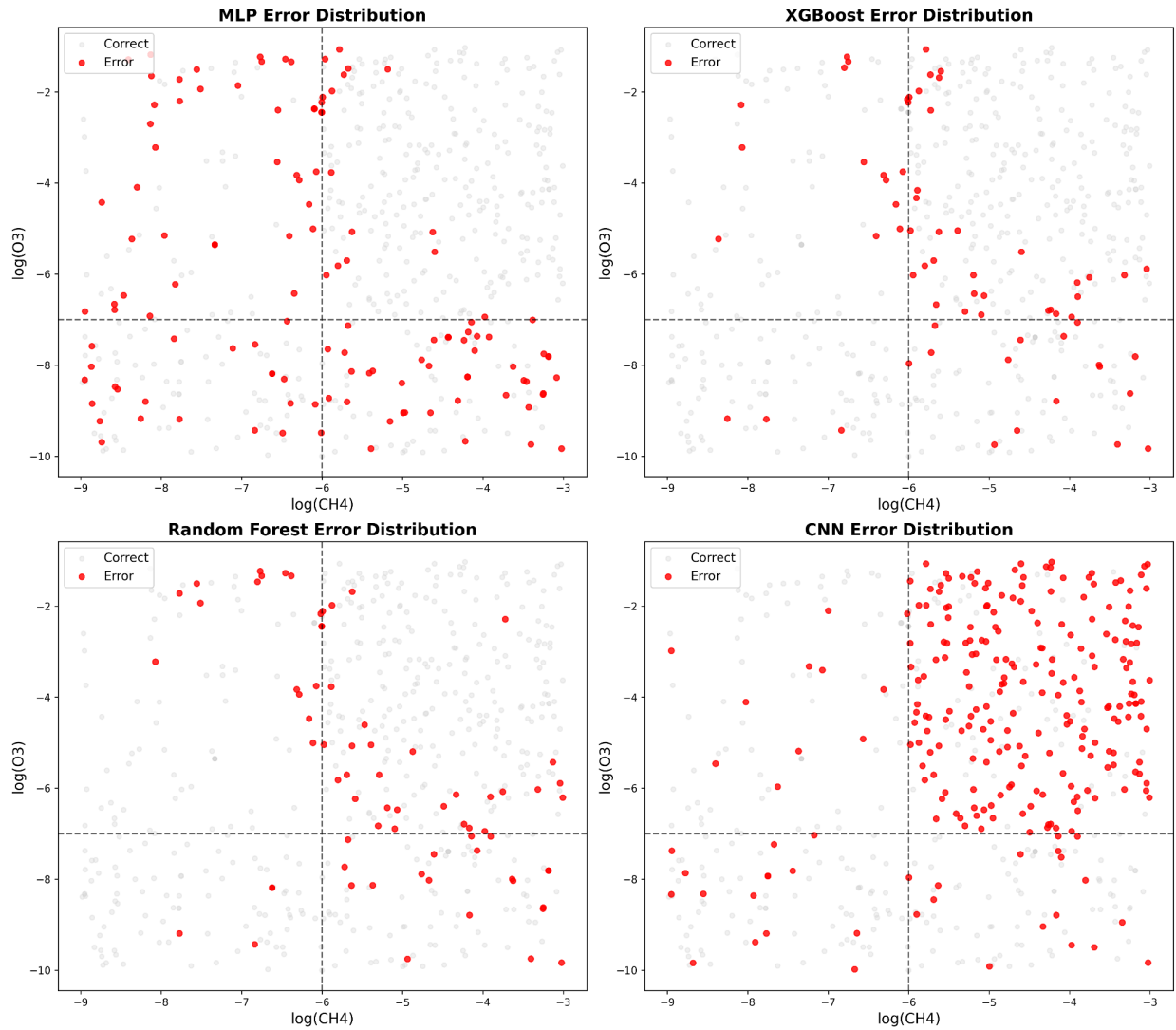


Figure 5: Distribution of classification errors relative to atmospheric mixing ratios of CH_4 and O_3 . Misclassifications (red) cluster heavily along the predefined biosignature detection thresholds of $\text{Log}(CH_4) = -6.0$ and $\text{Log}(O_3) = -7.0$. This pattern highlights the inherent ambiguity in marginal detections, where the amplitude of molecular absorption features is comparable to the stochastic noise injected into the synthetic spectra.

Model Limitations: Our current framework utilizes a binary classification approach, which is optimized for rapid target filtering but lacks the quantitative depth of traditional retrieval methods. Unlike Bayesian frameworks that provide full posterior probability distributions and chemical abundance estimates (35), our models provide a discrete label that cannot convey statistically rigorous confidence intervals for individual molecular species. Furthermore, the reliance on a fixed biosignature pair (CH_4/O_3) may be susceptible to "false-positive" biosignatures. For instance, high-metallicity atmospheres can sometimes exhibit spectral features that mimic the flat baseline of cloudy terrestrial worlds, potentially leading to misclassifications in the absence of a multi-label regression framework (36).

Simulation Limitations: A primary limitation of this study is the reliance on synthetic data generated via TauREx 3, which assumes a 1D, spherically symmetric atmospheric profile. Real exoplanetary atmospheres are inherently 3D environments characterized by complex day-night temperature gradients, inhomogeneous cloud coverage, and photochemical hazes (37). These 3D effects introduce spectral asymmetries and systematic biases that are not captured in 1D radiative transfer models. While the injection of Gaussian noise ($\text{SNR} = 15$) attempts to replicate Ariel's observational constraints, the lack of unmodeled 3D atmospheric dynamics means our models may overestimate performance when applied to real, heterogeneous exoplanetary data (38).

Future work can incorporate transfer learning to bridge the gap between simulated and real datasets. Transfer learning (39) is a method that involves pretraining models on one dataset and fine-tuning them on another. In the context of this work, transfer learning could be incorporated by first training on real low-resolution datasets, then adapting to the synthetic HR spectra that we used. This could improve both robustness and generalization. A transfer learning framework would allow models to learn realistic noise characteristics from low resolution spectra, then pivot to high resolution spectra that would tune it for richer chemical detail and allow it to be employed in higher tiers of the Ariel mission.

4.4 Probability Calibration and Mission Triage Reliability

As established, the ultimate goal of developing these machine learning models is to inform automated triage for missions such as Ariel. Ariel will employ a hierarchical observational strategy to optimize telescope time: Tier 1 conducts a broad reconnaissance of roughly 1,000 planets, Tier 2 provides a deeper chemical survey for a subset of approximately 500 atmospheres, and Tiers 3 and 4 are strictly reserved for repeated, high-resolution benchmark observations and phase curves of the most exceptional targets (5, 40). Tier 3 provides high-resolution ($R \approx 100\text{-}200$) benchmark observations for a select sample of 50 to 100 planets. Because both Tier 4 telescope time and the supercomputing hours required for full Tier 3 Bayesian atmospheric retrievals are finite resources, it is critical to aggressively prioritize the high-resolution Tier 3 catalog. When selecting a machine learning model to serve as this intra-Tier 3 prioritization filter, the decision depends heavily on the operational cost of classification errors.

When evaluated across the five independent test sets, the two tree-based ensembles—XGBoost (88.17%) and Random Forest (86.51%)—emerged as the top two models in overall predictive accuracy. Their operational reliability, however, diverges. As a primary measure of calibration we use the reliability diagram (Figure 6), which compares predicted confidence against empirical outcome frequency independently of raw accuracy. XGBoost's curve maintains near-diagonal alignment across the probability range, indicating that its confidence estimates track observed occurrence rates closely; for instance, predictions near 80% confidence correspond to an approximately 80% positive rate in the test set. The Random Forest curve shows compression toward 0.5, the MLP a sigmoid distortion characteristic of overconfidence, and the 1D-CNN no reliable alignment.

We report Brier scores (XGBoost 0.0861, Random Forest 0.1304, MLP 0.1466, 1D-CNN 0.2625) as a corroborating summary statistic, but interpret them with one important caveat. The Brier score is a strictly proper scoring rule that combines calibration (reliability) with discriminative power (resolution); a lower score can therefore reflect stronger discrimination rather than better calibration alone (41, 42). Because XGBoost is also our most accurate classifier, part of its Brier-score advantage is attributable to discrimination, not calibration per se, and the Brier ranking should not be read as a pure calibration ranking. The reliability diagram is the cleaner evidence: it shows that XGBoost's advantage is genuinely one of calibration—its curve is closest to the diagonal—and not merely a by-product of accuracy. The two lines of evidence agree, which is why we treat XGBoost's calibration as a robust empirical result on this dataset rather than an artifact of a single conflated metric.

The Random Forest's weaker calibration is consistent with a known property of bagging ensembles: averaging predictions from trees trained on distinct bootstrap resamples compresses ensemble probabilities toward the center, undervaluing high-confidence and overvaluing low-confidence detections (42). The neural networks fare worse still, the MLP tending toward overconfidence and the 1D-CNN failing to calibrate. Post-hoc methods such as Platt scaling or isotonic regression could narrow these gaps, so XGBoost's advantage is one of convenience and robustness rather than an insurmountable barrier; however, such calibrators add a separate fitting stage whose reliability depends on scarce held-out calibration data (isotonic regression in particular is prone to overfitting when that data is limited (42)). A model that is already well-calibrated avoids this added stage and its failure modes. We therefore adopt XGBoost as our primary recommendation—not because the alternatives cannot be calibrated, but because it delivers reliable probabilities without an additional calibration step. We emphasize that this is an empirical result on the present synthetic dataset, not a theoretical guarantee, since gradient-boosted trees are not inherently calibrated.

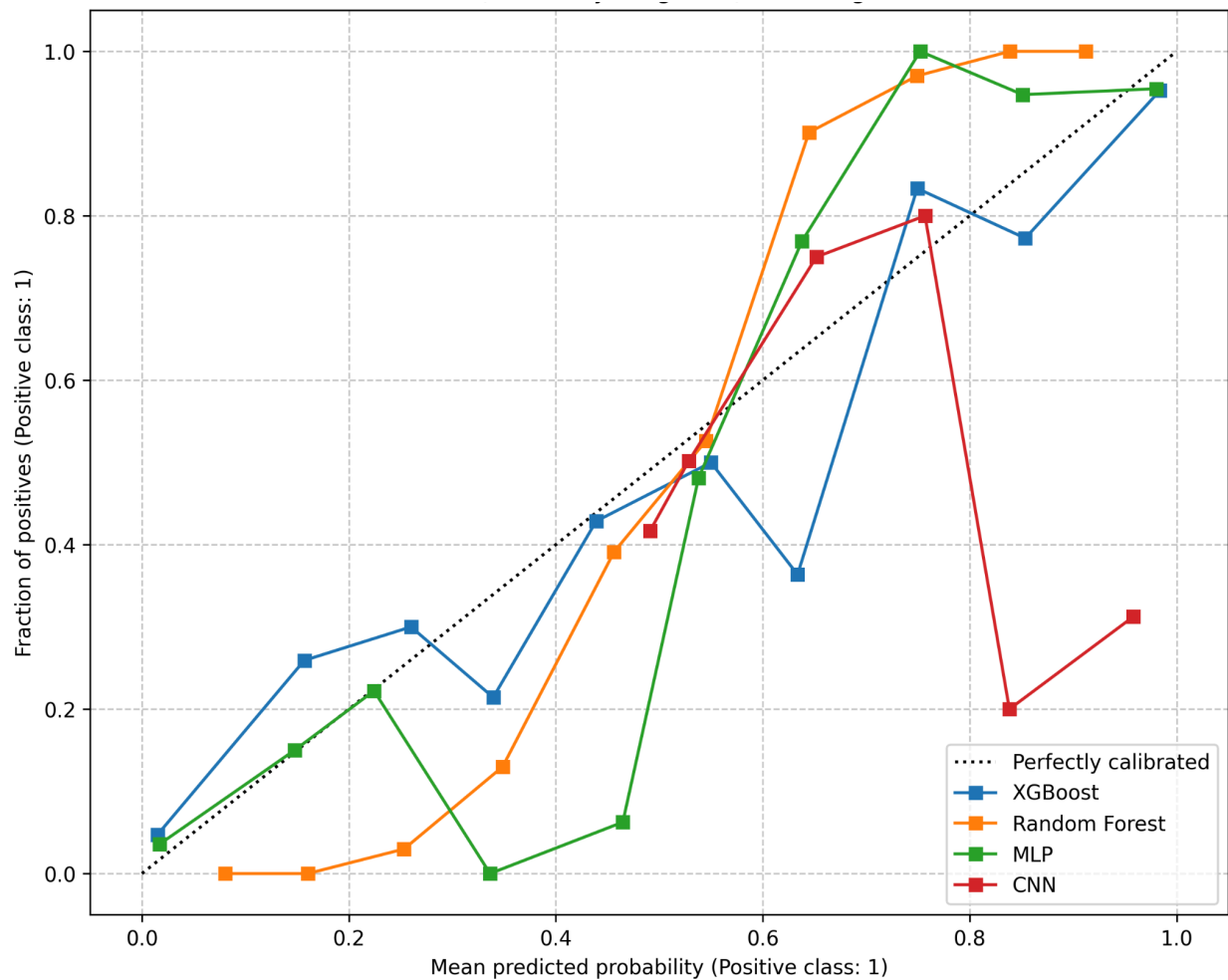


Figure 6. Calibration curves illustrating the reliability of model-predicted probabilities. A perfectly calibrated model follows the dotted diagonal line, indicating that predicted confidence matches the empirical occurrence rate. XGBoost (blue) demonstrates exceptional calibration (Brier score = 0.0861), rendering its probability estimates highly reliable for mission triage. In contrast, the MLP (green, Brier score = 0.1466) exhibits a sigmoid distortion indicative of overconfidence, while the Random Forest (orange, Brier score = 0.1304) exhibits vote compression indicative of underconfidence.

4.5 Application Trade-offs: Accuracy vs. Recall

While our models were optimized for balanced F1-performance during the tuning phase, the final deployment on the Ariel mission would likely involve threshold moving to align with specific scientific objectives. During an initial pass over the Tier 3 catalog, the classification threshold could be lowered to prioritize Recall, reducing the risk that a promising candidate is prematurely discarded. Conversely, when down-selecting the final few targets for the most resource-intensive follow-up—Tier 4 phase curves, ground-based ELT time, or full Bayesian retrievals—the threshold could be raised to maximize Precision, reserving those scarce resources for only the highest-probability candidates. This flexibility, combined with XGBoost's superior calibration, supports mission-specific risk management.

5. Conclusion

In this study, we evaluated the feasibility of using machine learning to classify exoplanet transmission spectra as biosignature or non-biosignature cases within an Ariel-like observational framework. By training and comparing four models—XGBoost, Random Forest, Multi-Layer Perceptron, and a 1D Convolutional Neural Network—on synthetic spectra generated with MultiREx and processed through a PCA-based feature extraction pipeline, we demonstrated that supervised learning methods can effectively distinguish between chemically distinct atmospheric regimes using high-dimensional spectral data.

Among the evaluated approaches, the tree-based ensembles (XGBoost and Random Forest) substantially outperformed the deep neural networks. Because their node-splitting is invariant to per-feature scale, the tree models exploited the chemically informative low-variance components natively, whereas the neural networks required explicit whitening of the PCA features even to remain competitive and still fell short. However, beyond aggregate accuracy metrics, this study highlights the importance of evaluating models within an operational calibration context. While Random Forest and XGBoost both provide robust, scale-invariant feature extraction, XGBoost's sequential boosting framework yields substantially better probability calibration (0.0861 Brier score) than Random Forest's compressed bagging averages (0.1304). Consequently, XGBoost is our primary triage recommendation for next-generation space surveys.

A central finding of this work is the critical role of feature engineering in extracting chemically meaningful signals from transmission spectra. The application of PCA enabled dimensionality reduction while preserving both broad continuum structure and high-frequency absorption features. Although PC0 and PC1 captured most of the broad physical variance, later components preserved chemically informative spectral structure, allowing the models to classify biosignature and non-biosignature cases effectively.

Beyond aggregate performance metrics, this study emphasizes the importance of evaluating machine learning models within a physical context. Error analysis revealed that model failures are not random, but instead cluster in well-defined regions of parameter space, particularly for planets with low atmospheric scale heights or near biosignature detection thresholds. These regimes correspond to physically challenging observational conditions, where spectral features are intrinsically weak or comparable to the noise level. This alignment between model behavior and known atmospheric physics provides additional confidence that the models are learning meaningful patterns rather than spurious correlations.

Despite these promising results, several limitations remain. The binary classification framework simplifies the complexity of atmospheric characterization and does not provide quantitative estimates of molecular abundances or uncertainties. Additionally, the reliance on synthetic spectra generated under 1D atmospheric assumptions limits the direct applicability of these models to real exoplanet observations, which may exhibit multidimensional structure, clouds,

and photochemical effects not captured in the simulations. These factors may lead to optimistic performance estimates when transitioning to real-world data.

Future work can address these limitations by incorporating more physically realistic atmospheric models, expanding to multi-label or regression-based approaches, and exploring transfer learning techniques to bridge the gap between synthetic and observed spectra. Additionally, further interpretability analysis—such as linking model decisions directly to specific spectral regions—would strengthen confidence in the physical validity of machine learning predictions.

Overall, this work demonstrates that machine learning, when combined with physically informed preprocessing and evaluation, can serve as an effective tool for prioritizing exoplanet targets in large-scale survey missions. As observational datasets continue to expand in both size and complexity, such approaches will play an increasingly important role in guiding the search for potentially habitable worlds.

Data and Code Availability Statement

The synthetic spectroscopic datasets, the five independent testing sets, and the complete source code for hyperparameter optimization, master evaluations, statistical McNemar testing, and physical/chemical visualizations are open-source and publicly available. The code repository can be accessed at <https://github.com/oy2017/BioSignatureDetectionModel>. All underlying simulated radiative transfer physics can be reproduced using the public MultiREx and TauREx 3 libraries.

References

1. NASA. (n.d.-a). *Exoplanet Exploration: Planets Beyond our Solar System*. National Aeronautics and Space Administration.
2. NASA. (n.d.-b). *The Search for Life*. National Aeronautics and Space Administration.
3. Schwieterman, E. W., Kiang, N. Y., Parenteau, C. E., Harman, C. E., Dassarma, S., Fisher, T. M., ... & Lyons, T. W. (2018). Exoplanet biosignatures: A review of remotely detectable signs of life. *Astrobiology*, 18(6), 663-708.
<https://doi.org/10.1089/ast.2017.1729>
4. Kiang, N. Y., Domagal-Goldman, S., Parenteau, M. N., Catling, D. C., Fujii, Y., Meadows, V. S., ... & Walker, S. I. (2018). Exoplanet biosignatures: At the dawn of a new era of planetary observations. *Astrobiology*, 18(6), 619-629.
5. Tinetti, G., Drossart, P., Eccleston, P., Hartogh, P., Heske, A., Leconte, J., ... & Zapata, J. (2018). A chemical survey of exoplanets with ARIEL. *Experimental Astronomy*, 46, 135-209.
6. European Space Agency. (2020). *Ariel definition study report (Red Book)*. ESA/SCI(2020)1.
7. Edwards, B., & Tinetti, G. (2022). The Ariel mission reference sample. *The Astronomical Journal*, 164(1), 15.

8. Madhusudhan, N. (2019). Exoplanetary atmospheres: Key insights, challenges, and prospects. *Annual Review of Astronomy and Astrophysics*, 57, 617-663.
9. Snellen, I. A., de Kok, R. J., de Mooij, E. J., & Albrecht, S. (2010). The orbital motion, absolute mass and high-altitude winds of exoplanet HD 209458b. *Nature*, 465(7301), 1049-1051.
10. Fujii, Y., Angerhausen, D., Deitrick, R., Domagal-Goldman, S., Grenfell, J. L., Hori, Y., ... & Pallé, E. (2018). Exoplanet biosignatures: Observational prospects. *Astrobiology*, 18(6), 739-778.
11. Brogi, M., & Line, M. R. (2019). Retrieving temperatures and abundances of exoplanet atmospheres with high-resolution cross-correlation spectroscopy. *The Astronomical Journal*, 157(3), 114.
12. NASA Exoplanet Archive. Atmospheric Spectroscopy Table. NASA Exoplanet Science Institute / Caltech-IPAC.
13. Sing, D. K., Fortney, J. J., Nikolov, N., Wakeford, H. R., Kataria, T., Evans, T. M., ... & Vidal-Madjar, A. (2016). A continuum from clear to cloudy hot-Jupiter exoplanets without primordial water depletion. *Nature*, 529(7584), 59-62.
14. Wordsworth, R., & Kreidberg, L. (2022). Atmospheres of rocky exoplanets. *Annual Review of Astronomy and Astrophysics*, 60, 159-201.
15. Changeat, Q., & Yip, K. H. (2023). ESA-Ariel Data Challenge NeurIPS 2022: Introduction to exo-atmospheric studies and presentation of the Ariel Big Challenge (ABC). *Proceedings of Machine Learning Research*.
16. Yip, K. H., et al. (2023). Lessons Learned from Ariel Data Challenge 2022: Inferring Physical Properties of Exoplanets From Next-Generation Telescopes. *Proceedings of Machine Learning Research*.
17. Duque-Castaño, D. S., Zuluaga, J. I., & Flor-Torres, L. (2024). Machine-assisted classification of potential biosignatures in Earth-like exoplanets using low signal-to-noise ratio transmission spectra. *Monthly Notices of the Royal Astronomical Society*, 539(2). <https://doi.org/10.48550/arxiv.2407.19167>
18. Al-Refaie, A. F., Changeat, Q., Waldmann, I. P., & Tinetti, G. (2021). TauREx III: A fast, dynamic, and extensible framework for retrievals. *The Astrophysical Journal*, 917(1), 37.
19. Márquez-Neila, P., Fisher, C., Sznitman, R., & Heng, K. (2018). Supervised machine learning for analysing spectra of exoplanetary atmospheres. *Nature Astronomy*, 2(9), 719-724.
20. Yip, K. H., Changeat, Q., Waldmann, I. P., & Tinetti, G. (2021). Into the deep: How interpretable are the predictions of deep learning models in exoplanet atmospheric retrievals?. *The Astronomical Journal*, 162(5), 195.
21. Tennyson, J., Yurchenko, S. N., Al-Refaie, A. F., Clark, V. H., Chubb, K. L., Conway, E. K., ... & Yurchenko, O. P. (2020). The 2020 release of the ExoMol database: Molecular line lists for exoplanet and other hot atmospheres. *Journal of Quantitative Spectroscopy and Radiative Transfer*, 255, 107228.
22. Changeat, Q., Al-Refaie, A., Mugnai, L. V., Edwards, B., Waldmann, I. P., Pascale, E., & Tinetti, G. (2020). Alfnoor: A retrieval simulation of the Ariel target list. *The Astronomical Journal*, 160(2), 80.

23. Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5-32.
24. Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785-794.
25. Unlu, Forestano, Matchev & Matcheva (2023). Reproducing Bayesian posterior distributions for exoplanet atmospheric parameter retrievals with a machine learning surrogate model. *arXiv preprint*, arXiv:2310.10521.
26. Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning*. MIT Press.
27. Garvin, E. O., Bonse, M. J., Hayoz, J., Cugno, G., Spiller, J., Patapis, P. A., ... & Quanz, S. P. (2024). Machine learning for exoplanet detection in high-contrast spectroscopy: Revealing exoplanets by leveraging hidden molecular signatures in cross-correlated spectra with convolutional neural networks. *arXiv preprint*, arXiv:2405.13469. <https://doi.org/10.48550/arXiv.2405.13469>
28. Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., ... & Duchesnay, E. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12(Oct), 2825-2830.
29. Kingma, D. P., & Ba, J. (2014). Adam: A method for stochastic optimization. *arXiv preprint*, arXiv:1412.6980.
30. He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 770-778.
31. Grinsztajn, L., Oyallon, E., & Varoquaux, G. (2022). Why do tree-based models still outperform deep learning on typical tabular data? *Advances in Neural Information Processing Systems*, 35, 507–520.
32. Shwartz-Ziv, R., & Armon, A. (2022). Tabular data: Deep learning is not all you need. *Information Fusion*, 81, 84–90. <https://doi.org/10.1016/j.inffus.2021.11.011>
33. McNemar, Q. (1947). Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2), 153-157.
34. Efron, B., & Tibshirani, R. J. (1993). *An Introduction to the Bootstrap*. Chapman & Hall/CRC.
35. Madhusudhan, N. (2018). Atmospheric retrieval of exoplanets. *Handbook of Exoplanets*, 1-28.
36. Line, M. R., & Parmentier, V. (2016). The influence of nonuniform cloud cover on transit transmission spectra. *The Astrophysical Journal*, 820(1), 78. [doi:10.3847/0004-637X/820/1/78](https://doi.org/10.3847/0004-637X/820/1/78)
37. Caldas, A., Leconte, J., Selsis, F., Waldmann, I. P., Bordé, P., Rocchetto, M., & Charnay, B. (2019). Effects of a fully 3D atmospheric structure on exoplanet transmission spectra: Retrieval biases due to day–night temperature gradients. *Astronomy & Astrophysics*, 623, A161.
38. Feng, Y. K., Line, M. R., Fortney, J. J., Stevenson, K. B., Bean, J., Kreidberg, L., & Parmentier, V. (2016). The impact of non-uniform thermal structure on the interpretation of exoplanet transmission spectra. *The Astrophysical Journal*, 829(1), 52.

39. Pan, S. J., & Yang, Q. (2009). A survey on transfer learning. *IEEE Transactions on Knowledge and Data Engineering*, 22(10), 1345-1359.
40. Edwards, Mugnai, Tinetti, Pascale, & Sarkar (2019). An updated study of potential targets for Ariel. *The Astronomical Journal*, 157(6), 242.
41. Brier, G. W. (1950). Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1), 1–3.
42. Niculescu-Mizil, A., & Caruana, R. (2005). Predicting good probabilities with supervised learning. *Proceedings of the 22nd International Conference on Machine Learning*, 625–632.